

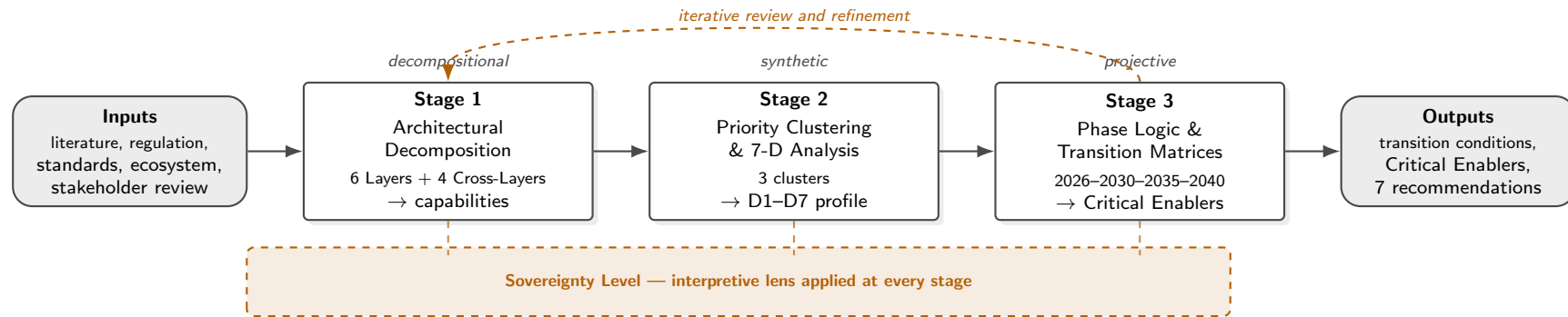


Figure 1: Three-stage methodology. Stage 1 decomposes the continuum into Security Layers, Transversal Cross-Layers, and constituent capabilities. Stage 2 consolidates capabilities into three priority clusters analysed through a seven-dimension template (D1–D7). Stage 3 projects the clusters onto three-phase transition matrices, terminating in a single Critical Enabler per phase boundary. The Sovereignty Level construct is applied as an interpretive lens across all stages rather than as a metric of any single stage.

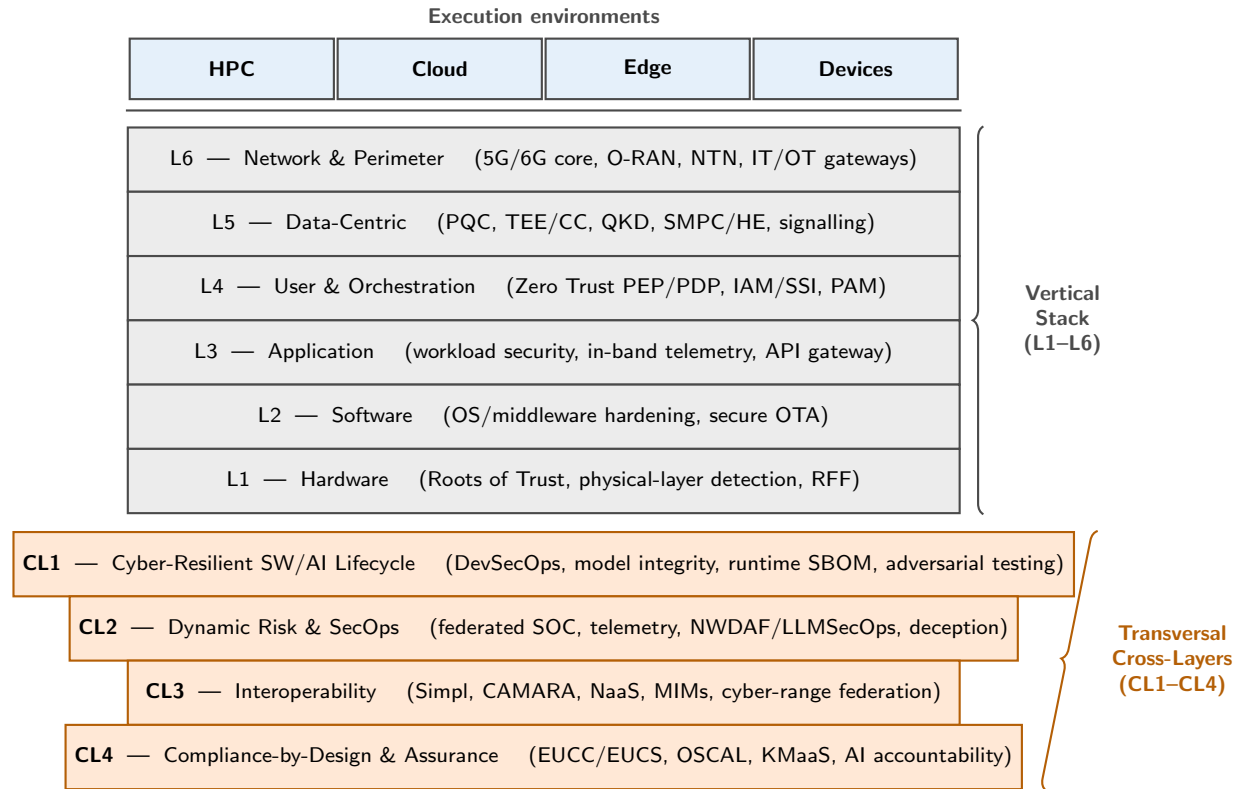


Figure 2: Security framework. Four execution environments (HPC, Cloud, Edge, Devices) host workloads governed by six Security Layers (L1–L6, vertical stack, lowest trust anchor at the bottom) and four Transversal Cross-Layers (CL1–CL4) acting as horizontal enablers spanning all environments and layers.

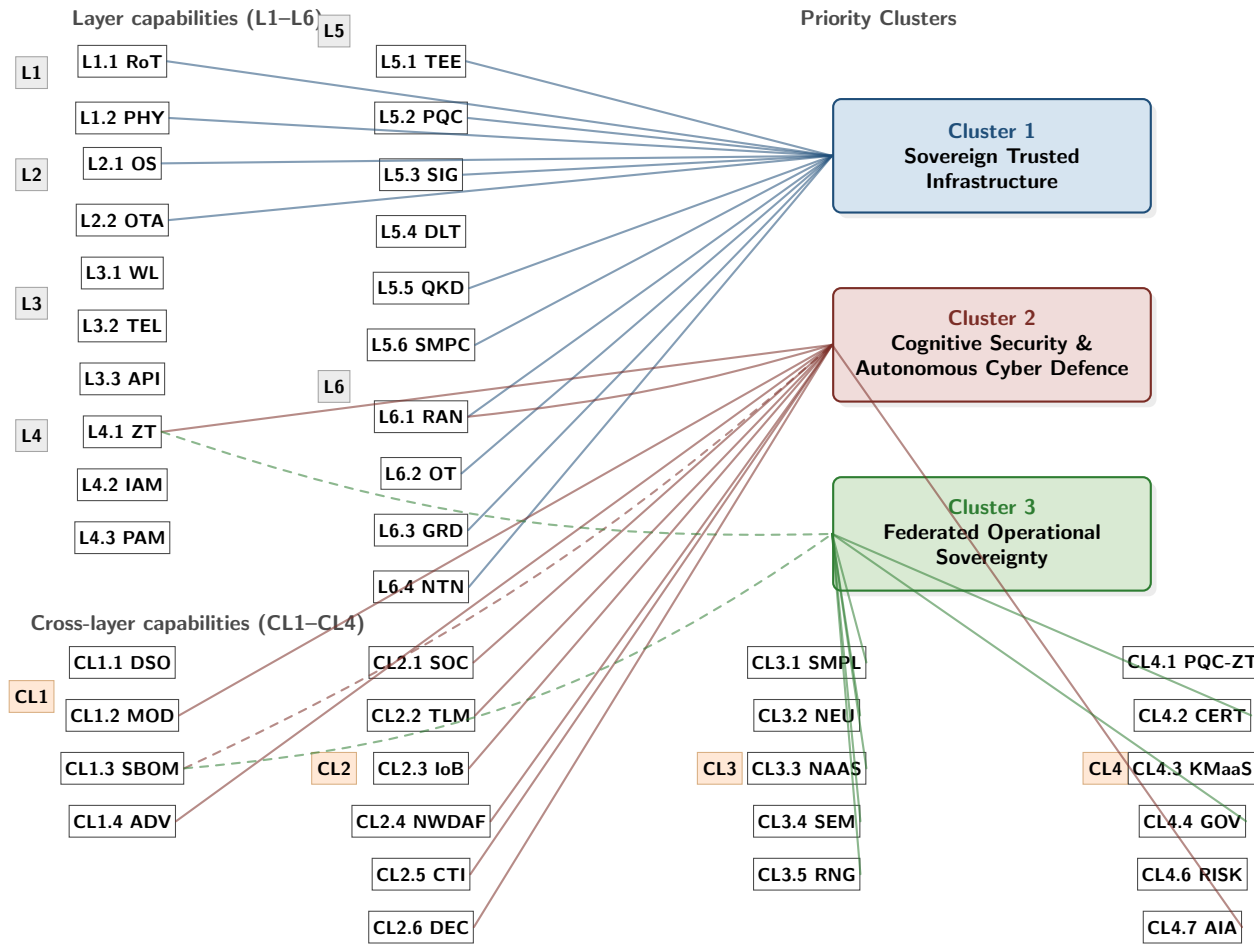


Figure 3: Stage 1 → Stage 2 capability aggregation. The 40 capabilities identified in Section ?? (left: 20 layer capabilities L1.1–L6.4; bottom: 20 cross-layer capabilities CL1.1–CL4.7) aggregate into the three priority clusters of Section ?. Three capabilities (L4.1 Zero-Trust orchestration, L6.1 Telco-cloud security, CL1.3 Runtime SBOM verification) are shared dependencies and act as bridges between clusters; Section ?? treats these shared capabilities as cross-cluster gate conditions.

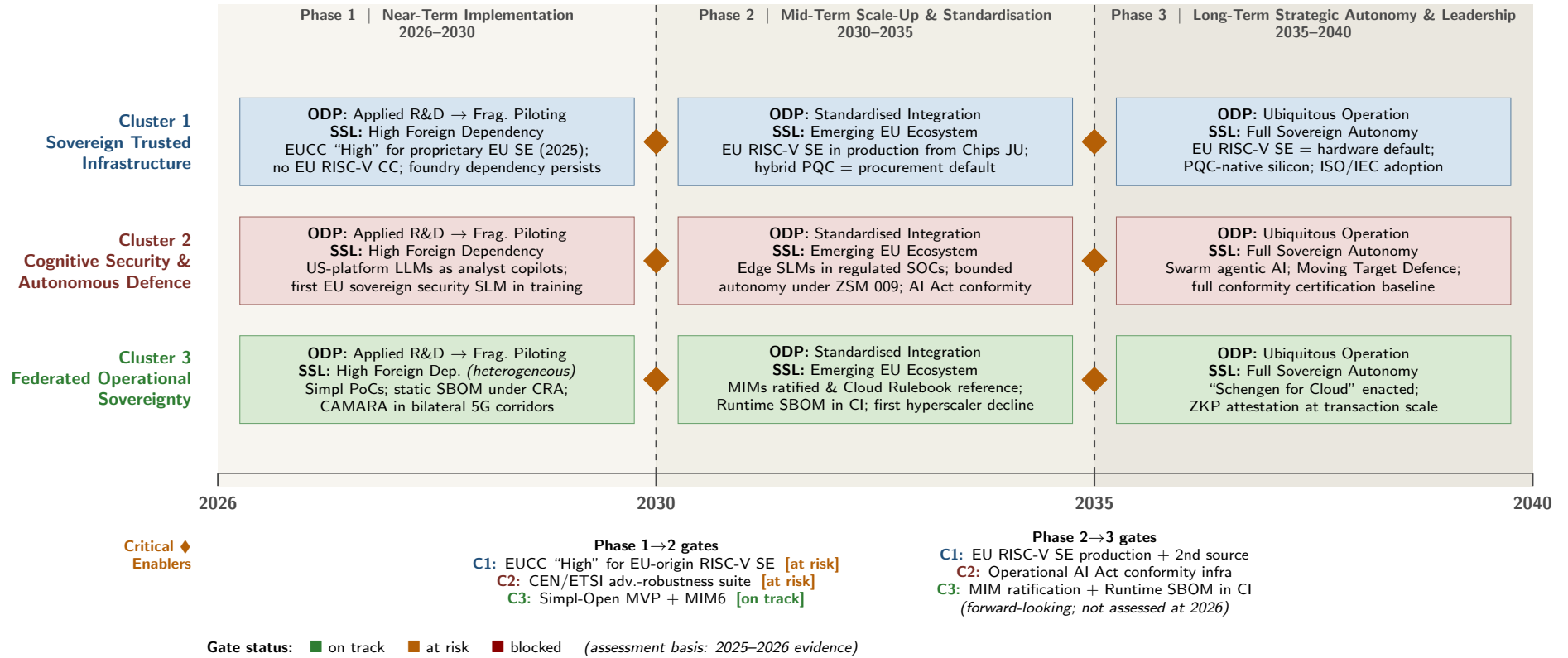


Figure 4: 2026–2040 phase roadmap. Each cluster (C1, C2, C3) is shown as a swim-lane with three phase columns; each cell records ODP and SSL bands plus a compressed Operational State. Diamond markers (◆) at phase boundaries denote the cluster-specific Critical Enabler. Phase 1→Phase 2 readiness, assessed against 2025–2026 evidence (Sections ?? and ??), is on track for Cluster 3 and at risk for Clusters 1 and 2.

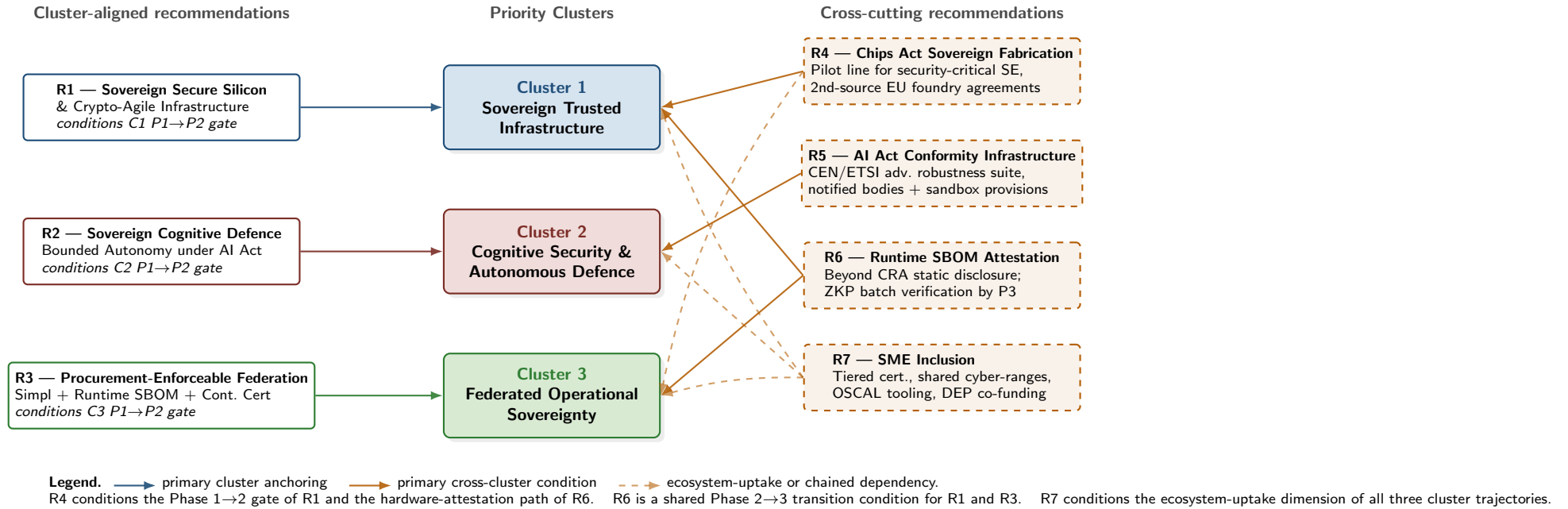


Figure 5: Recommendations × Clusters × Gates dependency map. Recommendations R1–R3 (left, solid borders) are cluster-aligned and condition the Phase 1→Phase 2 gate of their respective cluster. Recommendations R4–R7 (right, dashed borders) are cross-cutting: R4 conditions C1 and chains into C3 through R6; R5 conditions C2; R6 bridges C1 hardware attestation to C3 federation as a shared Phase 2→Phase 3 condition; R7 conditions the ecosystem-uptake dimension of all three cluster trajectories.